



Figure 1: GAE plug-in installation in Eclipse

available to the filesystem on App Engine

- Does not support domains without www such as <http://clean-clouds.com>
- Java applications may only use a subset of the classes from the JRE standard edition
- In-equality filters on two or more entity properties per query are not allowed in the datastore
The minimum request processing time is 60 seconds.

Data storage in GAE

Option	Description
GAE Datastore	• Distributed NoSQL service • Entities have a set of properties that support retrieval of entities with specific requirements • Schema-less entities • Python datastore and Java JDO/JPA interfaces are used for applying and enforcing the datastore structure within applications • Supports query engine and transactions • ACID transactions and efficient concurrency control
Google Cloud SQL	• Fully managed relational database service in Google's cloud • Capabilities of a popular MySQL database • Administration, management and monitoring of the database • Built-in back-up, recovery and patch management as it is managed by Google and not users • Built-in replication across multiple geographic regions • GUI to create, manage, configure and monitor your DB instance • Many options for configuration are available
Google Cloud Storage	• RESTful service for storing, managing and accessing data with Web-based interface • Reliable, highly available and scalable object store • Data replicated across multiple data centres • Read-your-write data consistency • Support to objects can be terabytes in size, resumable uploads and downloads • Auth 2.0 Authentication • Efficient access controls for individuals, projects and group-level • Authenticated uploads and downloads • Project, Buckets-Container and Objects-Data are basic building blocks of data

Language	Supported framework
Java	Spring Framework, Apache Struts 1 and Struts 2 with some changes, Servlet 2.5 technology using the open source Jetty Web Server, and JavaServer Faces with some changes. Java Runtime Environment uses Java 7. The limits of the sandbox environment are implemented in the JVM.
Python	Framework that supports the WSGI using the CGI adapter can be used to create an application on GAE. Frameworks like Django, GAE framework, CherryPy, web2py and webapp2, Flask, Pyramid, etc, include the Python standard library with limitations imposed by the sandbox environment. The webapp2 Python Web application framework is available for creating applications with ease.

SDK in GAE

App Engine provides a variety of services that help users to perform common management operations for an application such as URL Fetch, Mail, Memcache, Image Manipulation, etc. The GAE SDKs for Java, Python, PHP and Go each include a Web server application that emulates all of the GAE services such as APIs and libraries, secure sandbox environment, checks for attempts to access system resources on your local computer, tools to upload your application to GAE, etc. The Java SDK available as a Zip